Massive Dictionary with P columns
each column correspond to atoms
which represent spatial patterns

$$D = [D^{(1)} \mid D^{(2)} \mid D^{(3)} \mid \cdots \mid D^{(i)}]$$

Partitions groups
atoms that are
similar to each other

Cluster span
a low dimensional
subspace

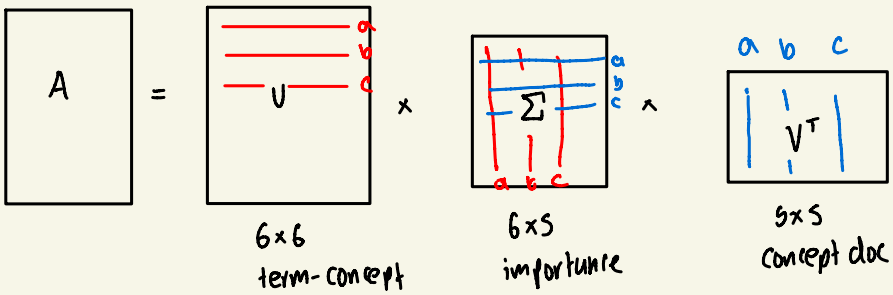
$$SVD = U \Sigma V^{T}$$



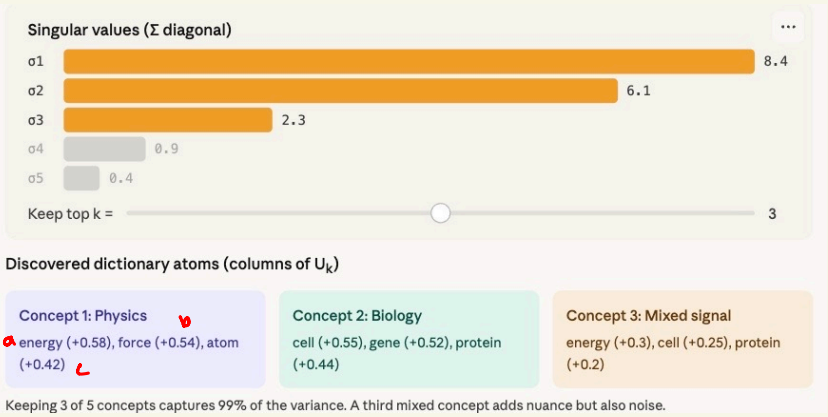
document

term

	Doc 1 Physics	Doc 2 Biology	Doc 3 Physics	Doc 4 Biology	Doc 5 Mixed
energy	3	0	2	0	1
force	2	0	3	0	1
cell	0	3	0	2	1
gene	0	2	0	3	0
atom	2	0	1	0	1
protein	0	1	0	2	1







$x_1 = (x_a, x_b)$
$x_2 = (x_c, x_d)$
$x_3 = (x_e, x_f)$

$\begin{bmatrix} x_a & x_c & x_e \\ x_b & x_d & x_f \end{bmatrix}$ $\begin{bmatrix} \vec{d_1} & d_2 \end{bmatrix}$ $x_2 = (0,2)$

$a(0.45, 0.89) \approx 0,2$

$x_a d_1 + x_b d_2$

$x_c d_1 + x_d d_2$

$x_e d_1 + x_f d_2$

$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \end{bmatrix}$ → start with signals from matrix

$D = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ → guess initialization

Update sparse function

$d_1 = (1,0)$
$d_2 = (1,1)$ } Formulae

① $D a = a_1 d_1 + a_2 d_2 = a_1 (1,0) + a_2 (1,1)$

$= (a_1 + a_2, a_2)$

$\therefore (a_1 + a_2, a_2) \approx (2,0)$

$a_1 d_1 + \overset{0}{a_2 d_2}$ ∴ only use d_1 atom since $a_2 = 0$

$a_2 = 0$
$\therefore a_1 = 2$

$a = (2,0)$ which perfectly represents P_1

② $(a_1 + a_2, a_2) \approx (0,2)$

$a_1 = -2$
$a_2 = 2$ } $x_2 \approx 0 d_1 + 2 d_2 = 2(1,1) = (2,2)$

L_1 penalty since $|2| + |-2| = 4$ } so $a_1 = 1$ $a_2 = 1$ $a(1,1)$ way better representation

③ $(a_1 + a_2, a_2) = (1,1)$

$a_2 = 1$

$a_1 + 1 = 1$ $a_1 = 0$

$D_a = 1 \cdot (1,1) = (1,1)$

$(1,1)$ d_2

Can perfectly represent it

since d_2 represents both a_1 and a_2 signals

Remove d_1 effect

$$d_1 a_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix} [2,0,0] = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$E_2 = X - d_1 a_1 = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 1 \end{bmatrix}$$

$$E_2^{(R1)} = \begin{bmatrix} 0 & 1 \\ 2 & 1 \end{bmatrix}$$

E: v_1: best direction in 2D → becomes d_2
v_1: How much each column uses it
σ_1: scaling

$d_2 \approx (0.45, 0.89)$

$$A \approx \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2.23 & 1.1 \end{bmatrix}$$

now try again

$x_1 = (2,0)$ (d_1) ✓ (d_2)

$x \cdot d_1 = (2 \cdot 1, 0 \cdot 0)$
$a_1 = (2,0)$
$(2,0) = 2(1,0)$

perfect fit

$x \cdot d_2 = 2(0.45) + 0(0.89)$
$a_1 = (0.9, 0)$
$0.9(0.45, 0.89) \approx (0.405, 0.801)$

bad fit penalty

Signal $x_2 = (0,2)$ (d_2) ✓

$x \cdot d_2 = 0(0.45) + 2(0.89) = 1.78$

$a_2 \approx (0, 1.78)$

(d_1)

$(0,2)$ cannot be represented by $(1,0)$

Signal $x_3 = (1,1)$ (d_2)

$x \cdot d_2 = (1(0.45) + 1(0.89) = 1.34$

$1.34 \cdot (0.45, 0.89) = (0.60, 1.19)$

(d_1)

$a_1 = 1$

$(1,1)$ ✗ bad one

utilising both

$(1,1) \approx a_1(1,0) + a_2(0.45, 0.89)$

$a_1 + 0.45a_2 = 1$

$0.89a_2 = 1$

$a_2 \approx 1.12$

$a_3 = (0.5, 1.12)$

$\approx (1,1)$

$a_1 + 0.45(1.12) = 1$

$a_1 = 0.5$

iterate till a converges

	coefficients	L_1 penalty	
d_2 only	$(0, 1.34)$	1.34λ	medium error
both	$(0.5, 1.12)$	1.62λ	lower error

Ways to make dictionary learning better

- Right number of atoms to prevent over fitting